

1

$$\frac{dy}{dx} = - \frac{3x^2 + y}{x + 6y^2} \Big|_{(1,1)} = - \frac{4}{7}$$

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = - \frac{(6+3) \cdot (x+6y^2) - (3x^2+y) \cdot (1+12y \cdot y')}{(x+6y^2)^2} \Big|_{(1,1)} = - \frac{430}{343}$$

2

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h} + \lim_{h \rightarrow 0} \frac{x f(x+h)}{h} = x^2 + 2$$

$$\therefore f'(x) = 6$$

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루팅이 중요한 것!!

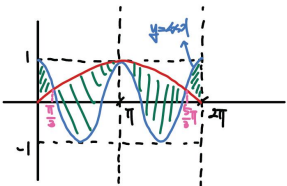
$$V = \pi r^2 h = 2\pi \quad \therefore h = \frac{2}{r^2}$$

$$S = 2\pi r h + \pi r^2 \xrightarrow{h = \frac{2}{r^2}} S = \pi r^2 + \frac{4\pi}{r}$$

$$S' = \frac{2\pi r^2 - 4\pi}{r^2} \quad r = \frac{2}{\sqrt{2}} \text{ 일때 } S \text{ 가 최소}$$

$$\therefore h = 2^{\frac{1}{2}} \Rightarrow r+h = 2^{\frac{3}{2}}$$

4



$$\begin{aligned} A &= 2 \int_0^{\pi/2} (\sin x - \sin \frac{x}{2}) dx + 2 \int_{\pi/2}^{\pi} (\sin \frac{x}{2} - \sin x) dx \\ &= 2 \left[-\cos x + 2\cos \frac{x}{2} \right]_0^{\pi/2} + 2 \left[-2\cos \frac{x}{2} - \cos x \right]_{\pi/2}^{\pi} \\ &= 2 \left(\left(\frac{6}{2} + \sqrt{3} \right) - (2) \right) - 2 \left((0) - \left(\sqrt{3} + \frac{3}{2} \right) \right) \\ &= 3\sqrt{3} - 4 + 3\sqrt{3} = 6\sqrt{3} - 4 \end{aligned}$$

5

$$(가) -x^2 = t \Rightarrow -dx = dt \quad \int_0^{\infty} t^2 e^{-t} dt : \text{수정}$$

$$(나) \lim_{x \rightarrow \infty} \frac{2+e^{-2x^2}}{1+\ln x} \xrightarrow{\text{L.C.T.}} \lim_{x \rightarrow \infty} \frac{-4x e^{-2x^2}}{\frac{1}{x}} = \lim_{x \rightarrow \infty} -4x^2 e^{-2x^2} = 0$$

$$(다) \int_0^{\pi/2} \frac{1}{x \sqrt{\sin x}} dx \geq \int_0^{\pi/2} \frac{1}{x^2} dx \quad \text{비교 판정법}$$

$$(라) \int_0^{\frac{1}{2}} \frac{-1}{\sqrt{3x-1}} dx + \int_{\frac{1}{2}}^{\frac{1}{3}} \frac{1}{\sqrt{3x-1}} dx = \lim_{x \rightarrow \frac{1}{3}^-} \left[-\frac{2}{3} (3x-1)^{\frac{3}{2}} \right]_0^{\frac{1}{2}} + \lim_{x \rightarrow \frac{1}{3}^+} \left[\frac{2}{3} (3x-1)^{\frac{3}{2}} \right]_{\frac{1}{2}}^{\frac{1}{3}} : \text{수정}$$

6

$$xt \leq u \Rightarrow xdt = du \Rightarrow \frac{1}{x} \int_0^{\infty} u e^{-u} du = \frac{1}{x^2} \quad \left[u(-e^{-u}) - (-e^{-u}) \right]_0^{\infty} = 1$$

$$\therefore f'(x) = -2x^{-3} \Big|_{x=2} = -\frac{1}{4}$$

7

$$\begin{aligned} f(x) &= \frac{4}{3} x \sqrt{4+x^2} = 2 \sqrt{4+x^2} \\ (4x)^p &= 1 + p \cdot \frac{p-1}{2!} x^2 + \dots \\ \Rightarrow f(x) &= \frac{8}{3} x \left(1 + \frac{1}{2} x^2 + \frac{(1/2)(-1/2)}{2!} x^4 + \frac{(1/2)(-1/2)(-3/2)}{3!} x^6 + \dots \right) \\ &= \frac{8}{3} \cdot \frac{1}{2} \cdot \frac{3}{8} \cdot \frac{1}{8} = \frac{1}{8} \cdot \frac{1}{2} \cdot \frac{3}{8} = \frac{3}{128} \end{aligned}$$

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$$\begin{aligned} f(x) &= (\sin x)^3 = \left(x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots \right)^3 \\ &\Rightarrow \left(x - \frac{x^3}{6} + \frac{x^5}{120} \right) \left(x - \frac{x^3}{6} + \frac{x^5}{120} \right) \left(x - \frac{x^3}{6} + \frac{x^5}{120} \right) \\ \text{항들:} & \quad \begin{array}{ccc} / & / & 5 \\ 3 & 3 & / \end{array} \quad \begin{array}{l} (x^3) \\ (x^3) \end{array} \\ \therefore 3 \cdot \left(\frac{1}{120} + \frac{1}{36} \right) &= \frac{1}{40} + \frac{1}{12} = \frac{3+10}{120} = \frac{13}{120} \\ \therefore f^{(10)}(0) &= \frac{13}{120} \cdot 10! = 13 \cdot 7 \cdot 6 = 546 \end{aligned}$$

9

$$Q_1 = \frac{1}{\frac{1}{2}} \cdot \frac{h\left(\frac{1}{2}\right)}{\frac{1}{2}} \cdot \frac{1}{2}$$

$$h\left(\frac{1}{2}\right) = \frac{1}{2} \Rightarrow \frac{1}{2} \cdot h = \int_1^2 \frac{h}{x} dx$$

$$h\left(\frac{1}{2}\right) \Rightarrow \frac{1}{2} dx = dt \quad : \int_0^{\frac{1}{2}} t dt = \frac{1}{2} \left(h\left(\frac{1}{2}\right)\right)^2$$

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(sol I)

$$y \text{ 증가} \Rightarrow \frac{dy}{dx} = \pm \infty$$

$$\frac{dy}{dx} = \frac{-\sin \theta \cdot \sin \theta + (1 + \cos \theta) \cos \theta}{-\sin \theta \cdot (1 + \cos \theta) - \sin \theta \cdot \cos \theta} = 0$$

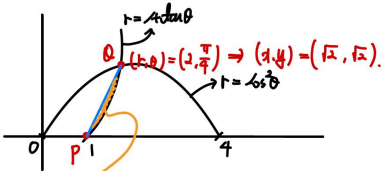
$$\Rightarrow 2 \sin \theta \cos \theta = -\sin \theta \Rightarrow \sin \theta (2 \cos \theta + 1) = 0$$

$$\therefore \theta = 0 \text{ } \times \text{ } \& \text{ } \frac{2}{3}\pi, \frac{4}{3}\pi \Rightarrow \frac{2}{3}\pi$$

(sol II)

$$r = 1 + \cos \theta \quad \text{graph} \quad \text{5월} \quad \therefore \frac{2}{3}\pi$$

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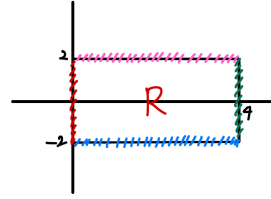


$$\frac{1}{2} \int_0^{\frac{\pi}{2}} (1 + \cos \theta)^2 d\theta - \Delta OPQ$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 + 2 \cos \theta + \cos^2 \theta) d\theta - \frac{1}{2} \cdot 1 \cdot \sqrt{2} = \frac{1}{2} [2\theta + 2\sin \theta + \frac{1}{2} \cos 2\theta]_0^{\frac{\pi}{2}} - \frac{\sqrt{2}}{2}$$

$$= \frac{1}{2} (4 \cdot \frac{\pi}{2}) - \frac{\sqrt{2}}{2}$$

12



i) 4부.

$$f = \langle x^2 - 2y - 4, -2x + 2y - 2 \rangle \quad \dots \quad x=3, y=1$$

$$\therefore f(3, 1) = 17 \quad \therefore \frac{2}{3}\pi$$

ii) 정제.

$$1. \quad x=0, -2 \leq y \leq 2 : f = 4y^2 - 2y + 24 \Rightarrow f_y = 8y - 2 = 0 \Rightarrow y = \frac{1}{4}$$

$$f\left(\frac{1}{4}\right) = \frac{95}{4}, \quad f(-2) = 44, \quad f(2) = 36$$

$$2. \quad y=-2, 0 \leq x \leq 4 : f = x^2 + 44 \Rightarrow f_x = 2x = 0 \Rightarrow x=0$$

$$f(0) = 44, \quad f(4) = 60$$

$$3. \quad x=4, -2 \leq y \leq 2 : f = 4y^2 - 8y + 24 \Rightarrow f_y = 8y - 8 = 0 \Rightarrow y = \frac{5}{4}$$

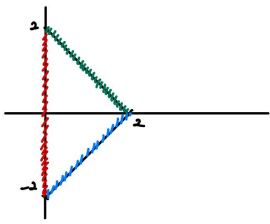
$$f\left(\frac{5}{4}\right) = \frac{11}{4}, \quad f(-2) = 60, \quad f(2) = 20$$

$$4. \quad y=2, 0 \leq x \leq 4 : f = x^2 - 8x + 36 \Rightarrow f_x = 2x - 8 = 0 \Rightarrow x=4$$

$$f(4) = 20, \quad f(0) = 36$$

$$\therefore M+M=27$$

13



i) 내부.

$$\vec{r} = \langle 2x-2, 2y \rangle = \vec{0} \quad \therefore \text{원점: } (1, 0).$$

$$f(1, 0) = -1$$

ii) 경계

① $x=0, -2 \leq y \leq 2$: $f = y^2 \Rightarrow f(0, 2) = 4, f(0, -2) = 4, f(0, 0) = 0$.

② $y=2-x, 0 \leq x \leq 2$: $f = x^2 - 2x + (x-2)^2 = x^2 - 4x + 4$.

$$\Rightarrow f' = 4x - 4 = 0 \quad \therefore x = 1$$

$$f(1) = -1, f(0) = 4, f(2) = 0$$

③ $y=x-2, 0 \leq x \leq 2$: $f = x^2 - 2x + (x-2)^2 = x^2 - 4x + 4$

$$\Rightarrow f' = 4x - 4 = 0 \quad \therefore x = 1$$

$$f(1) = -1, f(0) = 4, f(2) = 0$$

$$\therefore \text{i)와 ii)에 의해 } M = 4, m = -1 \Rightarrow M - m = 5 = b.$$

$$H_f = \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} \text{ \& } H_f(1, 0) > 0 \text{ \& } f_{xx} > 0 \quad \therefore \text{극대 / 계} = a.$$

$$\therefore a+b=6 //$$

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i) $\left| \frac{1}{3}, \frac{3}{1} \right| < 0$

ii) $\left| \frac{1}{3}, \frac{3}{1} \right| < 0$

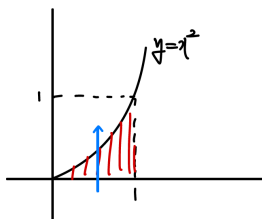
iii) $\left| \frac{3}{1}, \frac{1}{3} \right| > 0$ \& $f_{xx} > 0$. $\therefore$ 극소.

iv) $\left| \frac{3}{1}, \frac{1}{3} \right| < 0$

v) $\left| \frac{1}{3}, \frac{3}{1} \right| < 0$

$\Rightarrow$ 극대 / 계 //

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$$\Rightarrow \int_0^1 \int_0^{x^2} x \sin y \, dy \, dx$$

$$\begin{aligned} \Rightarrow \int_0^1 \left[x \sin y \right]_0^{x^2} dx &= \int_0^1 x \sin(x^2) dx = \left[-\frac{1}{2} \cos(x^2) \right]_0^1 \\ &= -\frac{1}{2} (\cos 1 - 1) = \frac{1}{2} (1 - \cos 1) // \end{aligned}$$

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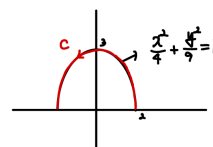
$$(x-1)^2 + y^2 = 1, \text{ \& } x=1, z=2x.$$

$$\begin{aligned} 2 \int_0^{\frac{\pi}{2}} \int_0^{2\cos\theta} r \, dr \, d\theta &= 2 \int_0^{\frac{\pi}{2}} \left[\frac{r^2}{2} \right]_0^{2\cos\theta} d\theta \\ &= 2 \int_0^{\frac{\pi}{2}} 2\cos^2\theta \, d\theta = \frac{16}{3} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{2} // \end{aligned}$$

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$$\Rightarrow \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{\cos\theta} \frac{r \cos\theta}{r^2} \cdot r \, dr \, d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\theta \, d\theta = 2 \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{2} //$$

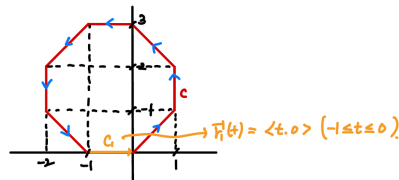
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$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_0^{\pi} \left\langle \frac{-3\sin t}{4\cos^2 t + 9\sin^2 t}, \frac{2\cos t}{4\cos^2 t + 9\sin^2 t} \right\rangle \cdot \langle -2\cos t, 2\sin t \rangle dt \\ &= 6 \int_0^{\pi} \frac{-3\sin t \cos t + 4\cos^2 t \sin t}{4\cos^2 t + 9\sin^2 t} dt. \end{aligned}$$

$$\begin{aligned} \frac{1}{9} \int_0^{\pi} \frac{1}{\frac{4}{9} + \tan^2 t} dt &= \frac{1}{9} \left(\frac{\frac{1}{2}}{\tan(\frac{\pi}{2})} \int_0^{\frac{\pi}{2}} \frac{9\sec^2 t}{\frac{4}{9} + \tan^2 t} dt + \frac{\frac{1}{2}}{\tan(\frac{\pi}{2})} \int_{\frac{\pi}{2}}^{\pi} \frac{9\sec^2 t}{\frac{4}{9} + \tan^2 t} dt \right) \\ &= \frac{1}{9} \cdot \left\{ \frac{\frac{1}{2}}{\tan(\frac{\pi}{2})} \left[\frac{3}{2} \ln \left(\frac{4}{9} + \tan^2 t \right) \right]_0^{\frac{\pi}{2}} + \frac{\frac{1}{2}}{\tan(\frac{\pi}{2})} \left[\frac{3}{2} \ln \left(\frac{4}{9} + \tan^2 t \right) \right]_{\frac{\pi}{2}}^{\pi} \right\} \\ &= \frac{1}{9} \cdot \frac{3}{2} \pi = \frac{\pi}{6} \quad \therefore 6 \int_0^{\pi} \frac{1}{4\cos^2 t + 9\sin^2 t} dt = \frac{\pi}{2} // \end{aligned}$$

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$$\int_{CVC_1} \vec{F} \cdot d\vec{r} = \iint_D (N_x - M_y) \, dA = 2 \cdot 7 = 14$$

$$\begin{aligned} \therefore \int_C \vec{F} \cdot d\vec{r} &= 14 - \int_{C_1} \vec{F} \cdot d\vec{r} = 14 - \int_{-1}^0 \langle e^t, t-1 \rangle \cdot \langle 1, 0 \rangle dt \\ &= 14 - [e^t]_{-1}^0 = 14 - (1 - e^{-1}) = 13 + \frac{1}{e} \end{aligned}$$

$$\therefore a=13, b=1 \Rightarrow 10a+b=131 //$$

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$$\begin{aligned}
 \stackrel{\text{Green}}{=} \iint_D (-2y) \, dA &= \int_0^1 \int_{x^2}^{\sqrt{x}} (-2y) \, dy \, dx = \int_0^1 (\sqrt{x} - x^2 - x + x^{\frac{5}{2}}) \, dx \\
 &= \frac{2}{3} - \frac{1}{3} - \frac{1}{2} + \frac{1}{5} = \underline{\underline{\frac{1}{30}}}
 \end{aligned}$$